

Final Year Project Documentation: IdeaForge Pakistan

PROJECT TITLE: IdeaForge Pakistan — AI-Powered Startup Validator

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DATE: April 2026

Chapter No 1: Introduction to the Problem

1.1 Introduction

IdeaForge Pakistan is a specialized web application designed to empower Pakistani entrepreneurs by providing institutional-grade validation for startup ideas. The platform leverages advanced Generative AI (LLMs) to perform deep market research, financial modeling, and strategic analysis tailored specifically to the local economic landscape of Pakistan.

1.2 Background

In Pakistan's burgeoning startup ecosystem, many founders fail due to a lack of data-driven market research and poor financial planning. Traditional consulting is expensive, and public market data is often fragmented. Founders frequently struggle to understand their Total Addressable Market (TAM) in PKR or identify local competitors across different cities like Karachi, Lahore, and Islamabad.

1.3 Purpose

The purpose of this project is to democratize startup consulting. By providing a scalable, AI-driven platform, IdeaForge Pakistan allows anyone with an idea to receive a comprehensive multi-page validation report in seconds, covering everything from revenue models to localized GTM (Go-To-Market) strategies.

1.4 Scope

- **AI Validation Engine:** Integration with Google Gemini for localized analysis.
- **Pakistani Market Context:** Support for PKR currency, local city data, and localized demand signals.
- **Documentation Export:** High-quality PDF generation for investor readiness.
- **Idea Management:** A secure dashboard to store and track the evolution of various startup concepts.

1.5 Objective

- To bridge the information gap for Pakistani entrepreneurs.
 - To provide realistic financial projections based on local CAC (Customer Acquisition Cost) and LTV (Lifetime Value).
 - To enable rapid iteration of business ideas through real-time AI feedback.
 - To produce professional reports suitable for incubators and early-stage investors.
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Chapter No 2: Software Requirement Specification (SRS)

2.1 Overall Description

The system is a cloud-based Web 3.0 application built on the Next.js framework. It acts as a bridge between user input (Business Ideas) and sophisticated AI analysis (Google Gemini).

2.1.1 Product Perspectives The application is a standalone platform but integrates with Supabase for authentication and database management, and Google's Generative AI for its core logic.

2.1.2 Product Features

- **User Authentication:** Secure sign-up/login via Supabase.
- **Dynamic Idea Input:** Smart forms to capture idea title, description, budget, and target region.
- **In-Depth Analysis:** 4-pillar analysis (Market, Financial, Strategic, and Executive).
- **Interactive Visuals:** Real-time charts for score breakdowns and dimension radars.
- **Heatmaps:** Localized market opportunity maps of Pakistan.
- **PDF Exporting:** Professional report generation.

2.1.3 Design and Implementation Constraints

- **Connectivity:** Requires active internet for AI processing and database sync.
- **API Limits:** Dependent on Gemini API rate limits and Supabase tier constraints.
- **Data Privacy:** Business ideas must be stored securely to prevent intellectual property theft.

2.1.4 Assumption and Dependencies

- Users have a basic understanding of their business concept.
- The AI model provides data accurate as of its last training update.

- Supabase provides 99.9% uptime for database and auth services.

2.2 System Features

2.2.1 System Feature 1: AI Result Generation The system takes raw text and transforms it into a structured JSON validation report using prompt engineering and Zod-based schema validation.

2.2.2 System Feature 2: Historical Tracking Users can revisit, delete, or regenerate past ideas from a centralized “History” page.

2.3 External Interface Requirements

2.3.1 User Interfaces

- Responsive Web Dashboard (Mobile-friendly).
- Interactive Recharts (Bar/Radar/Radial).
- Pakistan SVG Map with dynamic city-level scoring.

2.3.2 Hardware Interfaces The system is hardware-agnostic, requiring only a modern web browser on a PC or Smartphone.

2.3.3 Software Interfaces

- **Next.js 16+**: Frontend/Backend framework.
- **Supabase**: PostgreSQL database and Auth.
- **Gemini AI API**: Core intelligence engine.

2.3.4 Communications Interfaces

- HTTPS protocol for all data transmissions.
- RESTful API endpoints for internal communication.

2.4 Other Nonfunctional Requirements

2.4.1 Performance Requirements

- Analysis generation should complete within 15–20 seconds.
- Page transitions should be near-instant (under 2 seconds).

2.4.2 Safety Requirements

- No critical safety requirements as this is an information system.

2.4.3 Security Requirements

- Row Level Security (RLS) on Supabase ensures users can only see their own ideas.
- Environment variables for sensitive API keys.

Chapter No 3: Analysis (Use Case Model)

3.1 Identifying Actors and Use Cases

- **Primary Actor:** Entrepreneur (User)
- **Secondary Actor:** AI Engine (System)
- **Secondary Actor:** Database (Supabase)

3.2 Use Case Diagram

```
useCaseDiagram
    Entrepreneur --> (Sign Up / Login)
    Entrepreneur --> (Submit Idea)
    Entrepreneur --> (View Analytics)
    Entrepreneur --> (Download PDF Report)
    Entrepreneur --> (Manage History)
    (Submit Idea) ..> (AI Generation) : <<include>>
    (AI Generation) ..> (Database Save) : <<include>>
```

3.3 Event Flow for “Generate Report” Use Case

1. **Trigger:** User clicks “Analyze Idea”.
 2. **Step 1:** System validates form input.
 3. **Step 2:** System calls Gemini API with a system prompt optimized for Pakistan.
 4. **Step 3:** AI returns a structured JSON payload.
 5. **Step 4:** System saves the payload to PostgreSQL.
 6. **Step 5:** System redirects the user to the interactive results page.
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Chapter No 4: Design

4.1 Architecture Diagram

3-Tier Architecture: - **Presentation Layer:** Next.js Components (React 19). - **Application Layer:** Next.js Route Handlers (Edge Runtime). - **Data Layer:** Supabase (PostgreSQL) + Google Gemini AI.

4.2 ERD (Entity Relationship Diagram)

- **Profiles:** id, email, full_name.
- **Ideas:** id, user_id, title, description, region, status, result (JSONB).

4.3 Data Flow Diagram (DFD)

- **Level 0:** User -> [Input Ideas] -> IdeaForge System -> [Output Report] -> User.
 - **Level 1:** User -> Auth -> DB -> AI Processing -> Analysis Logic -> PDF Export.
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Chapter No 5: Implementation

5.1 Component Diagram

The system is divided into modular React components: - **Navbar** (Global) - **IdeaForm** (Submission) - **ScoreRing** (Visualization) - **PakistanMap** (Geographic data) - **IdeaResultsClient** (Main Logic)

5.3 Database Architecture (3-Tier)

1. **Database Server:** Supabase (PostgreSQL).
 2. **Application Server:** Next.js API Routes (Vercel).
 3. **Client:** Web Browser (React UI).
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Chapter No 6: Testing

6.1 Test Case Specification

ID	Test Case	Expected Result
TC1	Submission with empty title	“Missing required fields” alert
TC2	Successful AI Generation	JSON stored in DB & status marked “complete”
TC3	PDF Download trigger	Browser initiates download of .pdf file

6.2 Black Box Test Cases

- **Equivalence Class Partitioning:** Testing different city names (Karachi vs. Unknown City).
- **Boundary Value Analysis:** Description length (minimum 10 chars vs. maximum 2000 chars).

6.3 White Box Testing

- **Branch Coverage:** Ensuring both success and error paths in the `/api/validate` route are covered.

- **Statement Coverage:** Verifying that the Supabase update statement executes after every AI call.
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Chapter No 7: Tools and Technologies

7.1 Programming Languages

- **TypeScript:** Type-safe application logic.
- **JavaScript (ESNext):** Native browser functionality.
- **SQL:** Database queries.

7.2 Operating Environment

- **Framework:** Next.js 16.2.0 (Turbopack).
- **Hosting:** Vercel (Frontend & Serverless Functions).
- **Database:** Supabase (PostgreSQL 15).
- **AI Provider:** Google Gemini Pro (1.5 / 3.0 / Flash).
- **Styling:** Tailwind CSS 4.0.